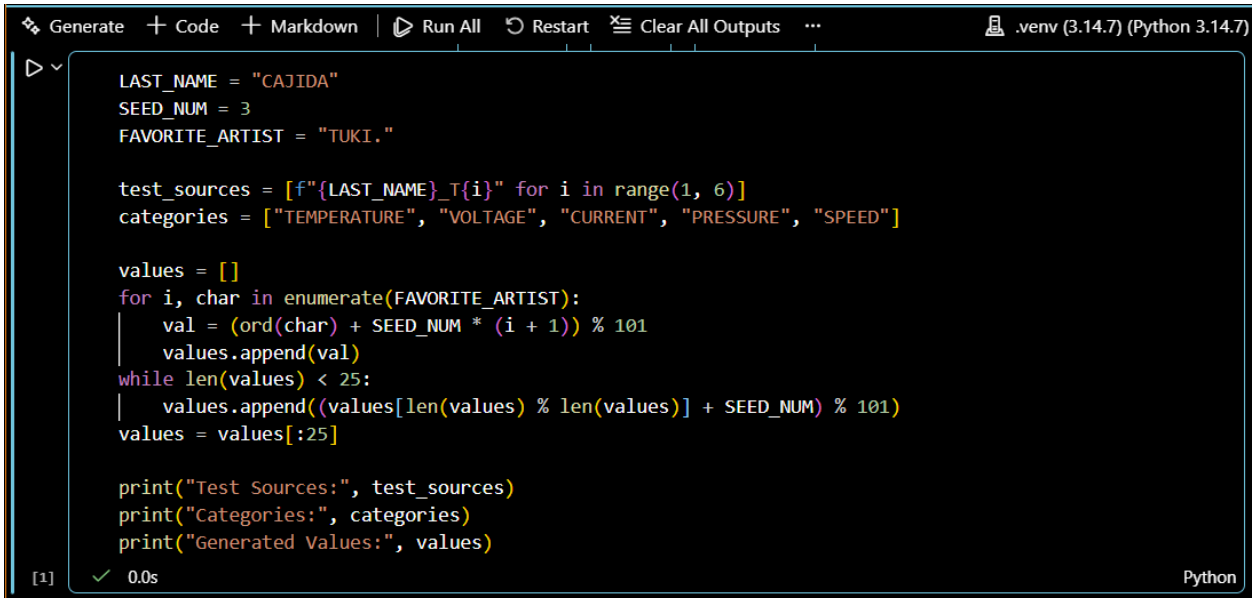
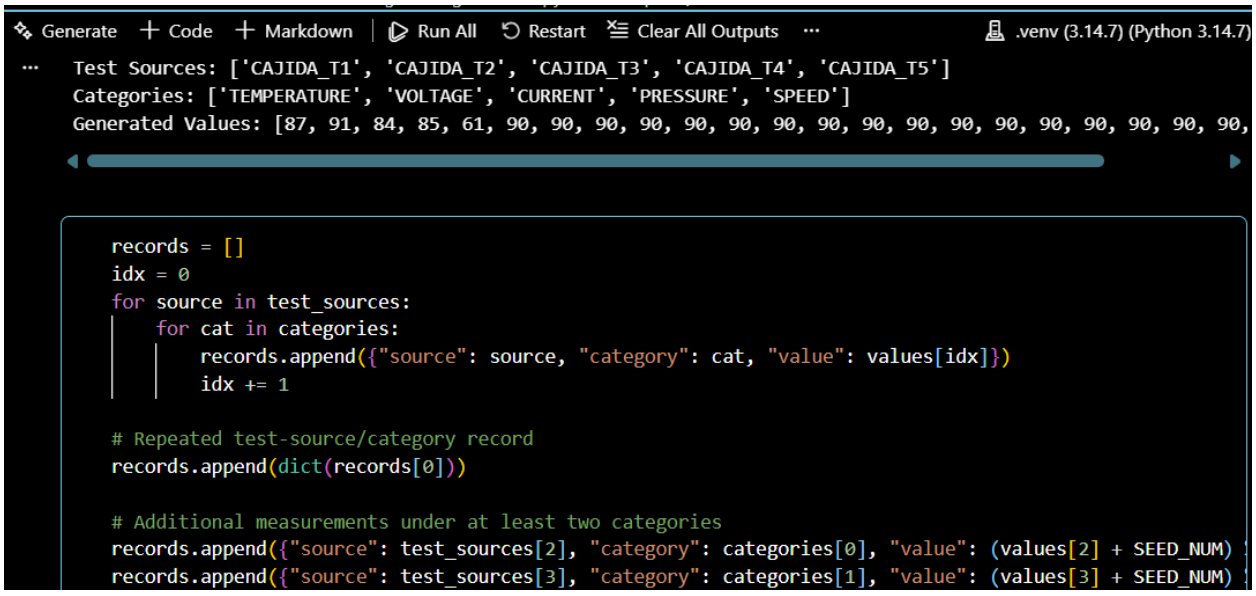
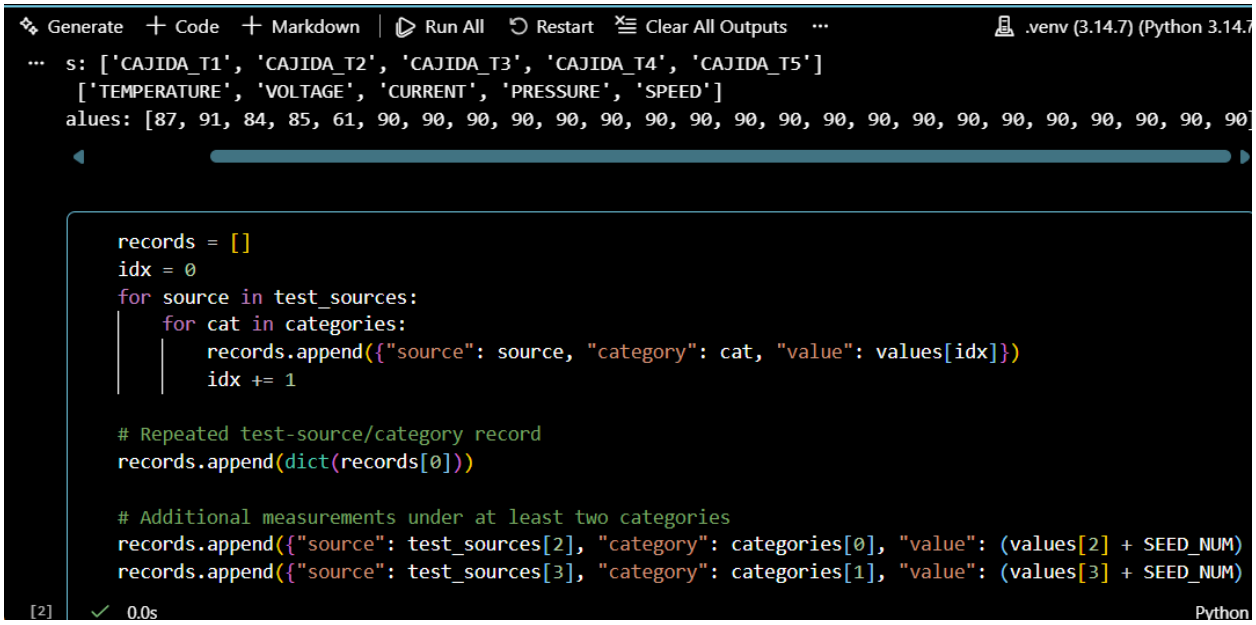
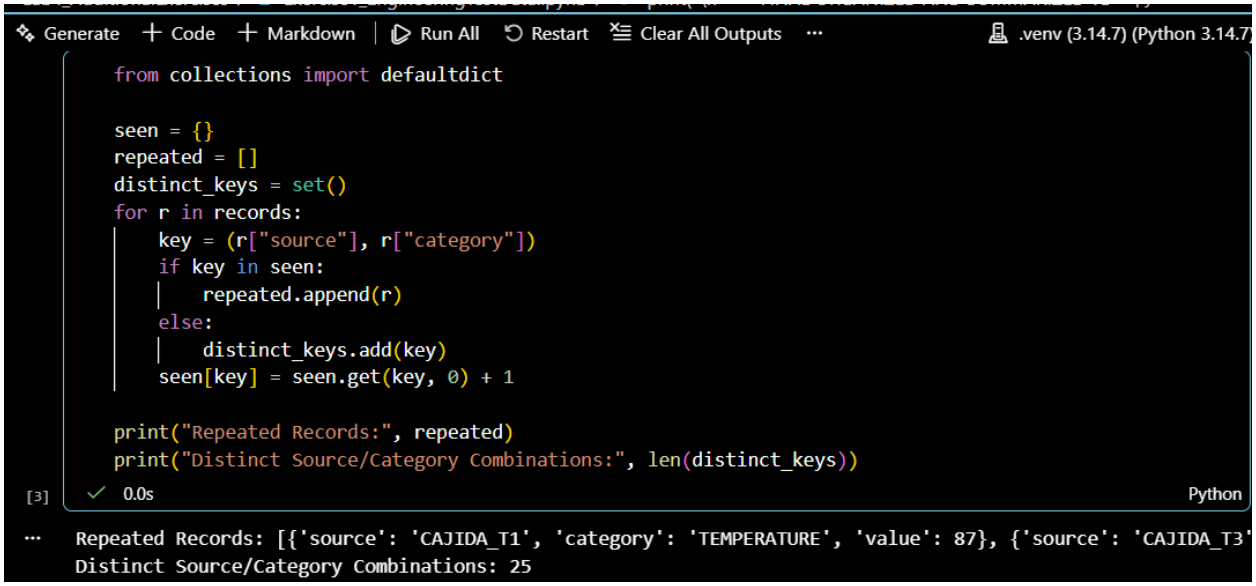

Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7)
LAST_NAME = "CAJIDA"
SEED_NUM = 3
FAVORITE_ARTIST = "TUKI."
test_sources = [f"{LAST_NAME}_T{i}" for i in range(1, 6)]
categories = ["TEMPERATURE", "VOLTAGE", "CURRENT", "PRESSURE", "SPEED"]
values = []
for i, char in enumerate(FAVORITE_ARTIST):
val = (ord(char) + SEED_NUM * (i + 1)) % 101
values.append(val)
while len(values) < 25:
values.append((values[len(values) % len(values)] + SEED_NUM) % 101)
values = values[:25]
print("Test Sources:", test_sources)
print("Categories:", categories)
print("Generated Values:", values)
[1]
0.0s
Python

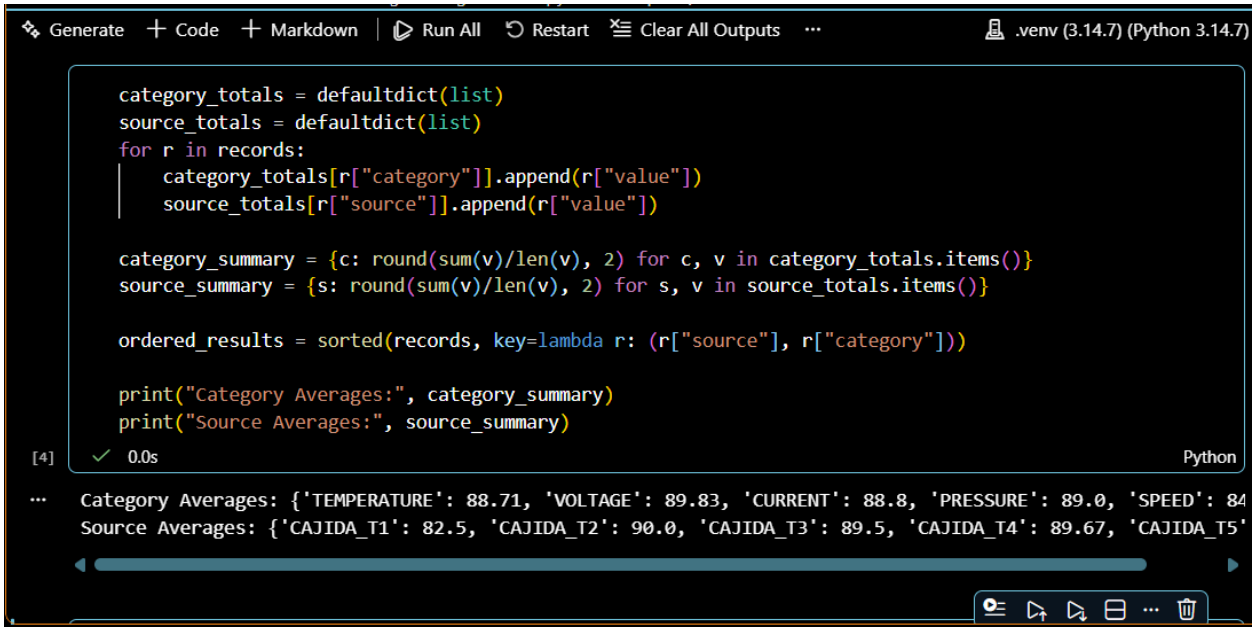

Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7)
Test Sources: ['CAJIDA_T1', 'CAJIDA_T2', 'CAJIDA_T3', 'CAJIDA_T4', 'CAJIDA_T5']
Categories: ['TEMPERATURE', 'VOLTAGE', 'CURRENT', 'PRESSURE', 'SPEED']
Generated Values: [87, 91, 84, 85, 61, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90,
records = []
idx = 0
for source in test_sources:
for cat in categories:
records.append({"source": source, "category": cat, "value": values[idx]})
idx += 1
Repeated test-source/category record
records.append(dict(records[0]))
Additional measurements under at least two categories
records.append({"source": test_sources[2], "category": categories[0], "value": (values[2] + SEED_NUM)
records.append({"source": test_sources[3], "category": categories[1], "value": (values[3] + SEED_NUM)

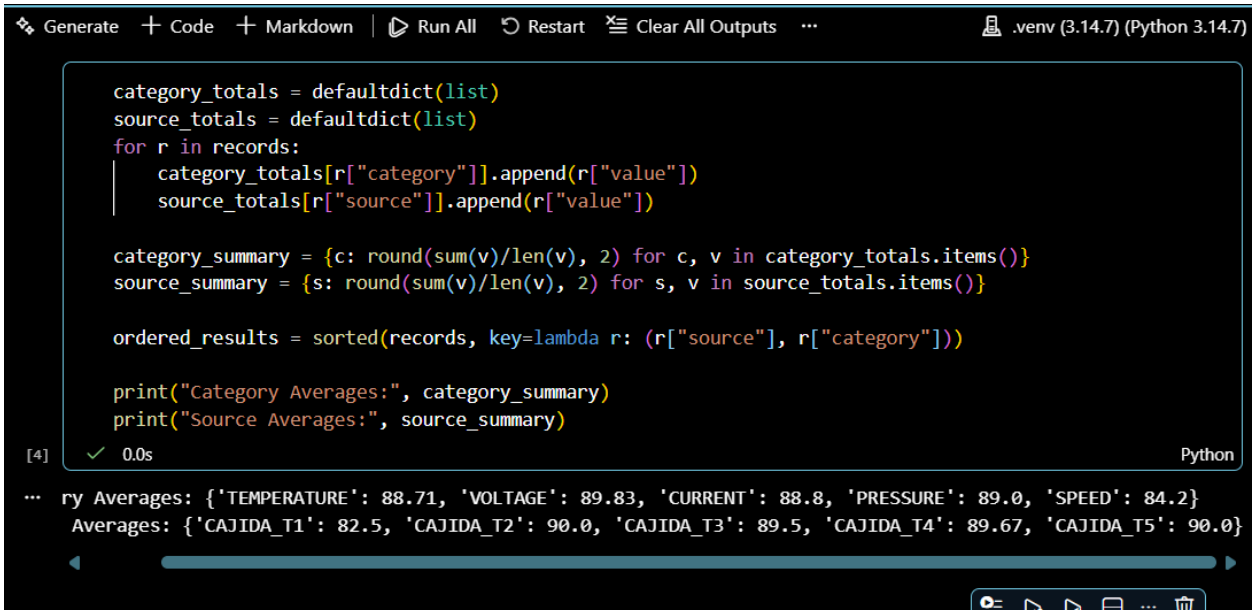

Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7
s: ['CAJIDA_T1', 'CAJIDA_T2', 'CAJIDA_T3', 'CAJIDA_T4', 'CAJIDA_T5']
['TEMPERATURE', 'VOLTAGE', 'CURRENT', 'PRESSURE', 'SPEED']
alues: [87, 91, 84, 85, 61, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90, 90]
records = []
idx = 0
for source in test_sources:
for cat in categories:
records.append({"source": source, "category": cat, "value": values[idx]})
idx += 1
Repeated test-source/category record
records.append(dict(records[0]))
Additional measurements under at least two categories
records.append({"source": test_sources[2], "category": categories[0], "value": (values[2] + SEED_NUM)
records.append({"source": test_sources[3], "category": categories[1], "value": (values[3] + SEED_NUM)
[2]
0.0s
Python


Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7)
records = []
idx = 0
for source in test_sources:
for cat in categories:
records.append({"source": source, "category": cat, "value": values[idx]})
idx += 1
Repeated test-source/category record
records.append(dict(records[0]))
Additional measurements under at least two categories
records.append({"source": test_sources[2], "category": categories[0], "value": (values[2] + SEED_NUM)
records.append({"source": test_sources[3], "category": categories[1], "value": (values[3] + SEED_NUM)
print(f"Total Records Generated: {len(records)}")
[2]
0.0s
Python
Total Records Generated: 28


Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7)
from collections import defaultdict
seen = {}
repeated = []
distinct_keys = set()
for r in records:
key = (r["source"], r["category"])
if key in seen:
repeated.append(r)
else:
distinct_keys.add(key)
seen[key] = seen.get(key, 0) + 1
print("Repeated Records:", repeated)
print("Distinct Source/Category Combinations:", len(distinct_keys))
[3]
0.0s
Python
Repeated Records: [{'source': 'CAJIDA_T1', 'category': 'TEMPERATURE', 'value': 87}, {'source': 'CAJIDA_T3'
Distinct Source/Category Combinations: 25


Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7
from collections import defaultdict
seen = {}
repeated = []
distinct_keys = set()
for r in records:
key = (r["source"], r["category"])
if key in seen:
repeated.append(r)
else:
distinct_keys.add(key)
seen[key] = seen.get(key, 0) + 1
print("Repeated Records:", repeated)
print("Distinct Source/Category Combinations:", len(distinct_keys))
[3]
0.0s
Python
_T3', 'category': 'TEMPERATURE', 'value': 87}, {'source': 'CAJIDA_T4', 'category': 'VOLTAGE', 'value': 88}]


Generate
Code
Markdown
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Clear All Outputs
.venv (3.14.7) (Python 3.14.7)
category_totals = defaultdict(list)
source_totals = defaultdict(list)
for r in records:
category_totals[r["category"]].append(r["value"])
source_totals[r["source"]].append(r["value"])
category_summary = {c: round(sum(v)/len(v), 2) for c, v in category_totals.items()}
source_summary = {s: round(sum(v)/len(v), 2) for s, v in source_totals.items()}
ordered_results = sorted(records, key=lambda r: (r["source"], r["category"]))
print("Category Averages:", category_summary)
print("Source Averages:", source_summary)
[4]
0.0s
Python
Category Averages: {'TEMPERATURE': 88.71, 'VOLTAGE': 89.83, 'CURRENT': 88.8, 'PRESSURE': 89.0, 'SPEED': 84
Source Averages: {'CAJIDA_T1': 82.5, 'CAJIDA_T2': 90.0, 'CAJIDA_T3': 89.5, 'CAJIDA_T4': 89.67, 'CAJIDA_T5'


Generate
Code
Markdown
Run All
Restart
Clear All Outputs
.venv (3.14.7) (Python 3.14.7)
category_totals = defaultdict(list)
source_totals = defaultdict(list)
for r in records:
category_totals[r["category"]].append(r["value"])
source_totals[r["source"]].append(r["value"])
category_summary = {c: round(sum(v)/len(v), 2) for c, v in category_totals.items()}
source_summary = {s: round(sum(v)/len(v), 2) for s, v in source_totals.items()}
ordered_results = sorted(records, key=lambda r: (r["source"], r["category"]))
print("Category Averages:", category_summary)
print("Source Averages:", source_summary)
[4]
0.0s
Python
ry Averages: {'TEMPERATURE': 88.71, 'VOLTAGE': 89.83, 'CURRENT': 88.8, 'PRESSURE': 89.0, 'SPEED': 84.2}
Averages: {'CAJIDA_T1': 82.5, 'CAJIDA_T2': 90.0, 'CAJIDA_T3': 89.5, 'CAJIDA_T4': 89.67, 'CAJIDA_T5': 90.0}